

ONLINE SHOPPING DATABASE

HACKATHON PROJECT

TEAM DETAILS

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Problem Statement

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ONLINE SHOPPING DATABASE

Design a relational database for an e-commerce system that manages customers, products, orders and order items.

Customers

Stored and managed in the database

Products

Stored and managed in the database

Orders

Stored and managed in the database

Order Items

Stored and managed in the database

Product Categories

Stored and managed in the database

Customer Order History

Stored and managed in the database

Problem / Need

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Need 1

Online stores handle large amounts of data.

Need 2

Customer and product details must be stored safely.

Need 3

Orders need to be tracked correctly.

Need 4

Product stock should be managed.

Need 5

Customers need easy access to their order history.

Proposed Solution

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SOLUTION

We develop a relational database to manage the complete shopping process.

Customer management

Manage customer

Product management

Manage product

Category management

Manage category

Order management

Manage order

Order item management

Manage order item

Order history

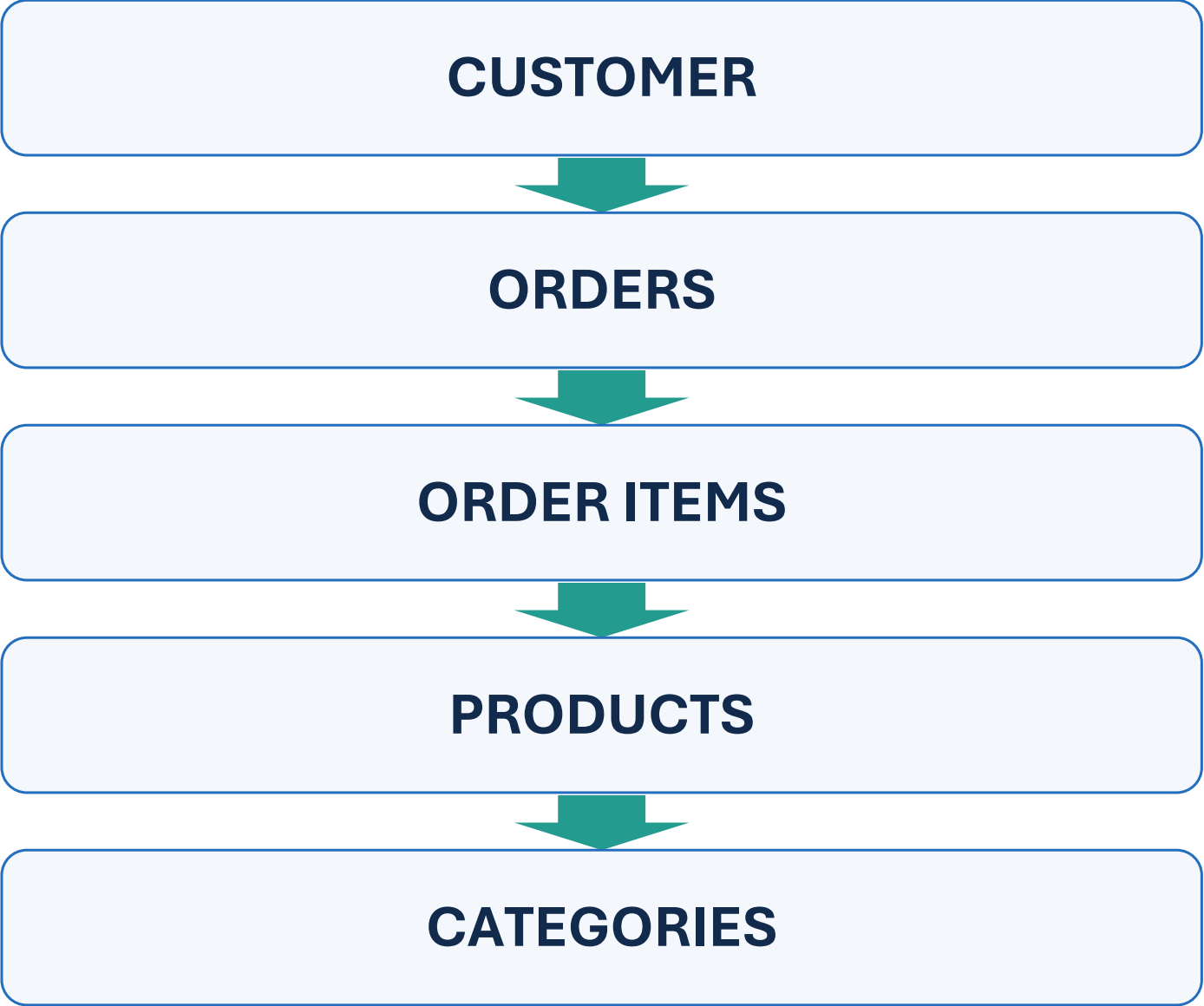
Manage order history

Database Tables

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TABLE	MAIN INFORMATION
Customer	Customer ID, name, email and phone
Product	Product ID, name, price and stock
Category	Category ID and category name
Orders	Order ID, customer ID, date and status
Order Items	Order ID, product ID, quantity and price
Order History	Previous orders placed by each customer

Database Structure



RELATIONSHIP

One customer → Many orders

RELATIONSHIP

One order → Many order items

RELATIONSHIP

One product → Many order items

RELATIONSHIP

One category → Many products

SQL Implementation

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```
CREATE DATABASE OnlineShopping;  
USE OnlineShopping;
```

```
CREATE TABLE Customer (  
  customer_id INT PRIMARY KEY,  
  name VARCHAR(50),  
  email VARCHAR(100),  
  phone VARCHAR(15)  
);
```

SQL OPERATIONS

```
CREATE  
INSERT  
SELECT  
UPDATE  
DELETE  
JOIN
```

Key Features

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✓ Customer registration

Feature of the shopping database

✓ Product categories

Feature of the shopping database

✓ Store order items

Feature of the shopping database

✓ Search product details

Feature of the shopping database

✓ Product listing

Feature of the shopping database

✓ Place orders

Feature of the shopping database

✓ View order history

Feature of the shopping database

✓ Manage product information

Feature of the shopping database

Advantages

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Reduces data duplication

Improves the database workflow

Easy data management

Improves the database workflow

Fast order tracking

Improves the database workflow

Maintains customer history

Improves the database workflow

Improves accuracy

Improves the database workflow

Easy to expand for an e-commerce application

Improves the database workflow

Future Scope

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Online payment integration

Future enhancement

Product search and filters

Future enhancement

Delivery tracking

Future enhancement

Sales and customer analytics

Future enhancement

Shopping cart

Future enhancement

Customer login

Future enhancement

AI-based product recommendations

Future enhancement

Conclusion

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ONLINE SHOPPING DATABASE

Provides a simple and organized way to manage customers, products and orders.

STORE

Store data securely

MANAGE

Manage orders easily

TRACK

Track customer history

RESULT

Support efficient online shopping

THANK YOU

ONLINE SHOPPING DATABASE

Hackathon Project

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